

DATA CLENASING GUIDELINES

The purpose of this document is to systematically generate consistent format and nomenclature for all MRO (Maintenance, Repairs and Operations) materials. These standards apply to all business units using the client specified ERP/EAM system and will be applied to any future additions to the material catalogue.

MATERIAL VALUES

- When material is referenced as a characteristic by itself, the values will be listed as follows:
PROCESS
BASE MATERIAL
SPECIFICATION
GRADE
EXAMPLE: GALV STEEL ASTM 105 GRADE B
EXAMPLE: COLD ROLLED STEEL ASTM 105 GRADE B
- The exception to the above mentioned arrangement is for Stainless Steels.
GENERAL ORDER OF STAINLESS STEELS ARE:
AISI
GRADE
BASE MATERIAL
SPECIFICATION
GRADE
EXAMPLE: 316 SS ASTM 182
- When more than one (1) material is given for a specific item, separate the materials with a slash.
EXAMPLE: STEEL/NITRILE
- O-RINGS: Use only the material value given in the source description
- Brand names may sometimes be used as a material value when they are a commercial description.
EXAMPLE: TEFLON or GRAFOIL

SIZE VALUES

- Descriptive value (i.e. MM, IN, FT) and specify the dimensional value (i.e. ID, OD, etc.) is used
- When referencing sizes that are not listed as individual characteristic, list the values in the following sequence: ID X OD X DIA X WD X LG X DP X HT X THK

Example 0.5IN ID X 0.75IN OD X 3IN LG

Example 0.5IN ID X 0.75IN OD X 0.25IN THK

- Specific orders of dimensions should be according to the sample values for each particular noun.
- Will always use a delimiter (e.g. DIA X ID X OD etc.) after the value when filling in the characteristic of SIZE.
- Multiple dimensions are represented using the space X space (X) between the measurements
EXAMPLE: 5 X 6, 2 X 3.25IN).
- GASKETS are normally sized by ID X OD X THK
EXAMPLE: 3IN ID X 4IN OD X 0.125IN THK
EXAMPLE: 75MM ID X 100MM OD X 6MM THK
- When gaskets have bolt holes, DIA will be used after the bolt hole size.
EXAMPLE: 0.5IN DIA BOLT HOLE
- O-RINGS are normally sized by the Inside Diameter and Width. An O-RING gauge should be used when determining O-RING sizes.
- NUT SIZES: Use nominal size only. Sometimes called thread size, tube size, pipe size, stud size, screw size, or bolt size. The diameter of a nut is the INSIDE DIAMETER, which is the same size as that of the bolt with which it is to be used.
- Dry Cell Batteries can be letter sizes (e.g., AA, C, D) can also be numeric. If quantities are needed use the (4) AA convention to indicate quantity.
- LARGE, MEDIUM, or SMALL are spelled out when describing sizes
- Quantities should always be enclosed with parenthesis ().
EXAMPLE: (8) 0.5IN DIA BOLT HOLES

TYPE VALUES

- Generally the second Modifier will be used as TYPE attribute
- If the more specific part is known, the value should indicate its closest level of hierarchy
- This excludes brand names or manufacturer types used to describe their products.

ELECTRIC RATING

- GENERAL ORDER OF ELECTRICAL VALUES: VOLTAGE, AMPERAGE, PHASE, HERTZ (CYCLES), WATTAGE, OTHERS (HORSEPOWER, OHM, VOLTAMP, ETC.)
EXAMPLE: 120VAC 15A 1PH 50/60HZ 6W 75HP
- When multiple electrical values are used together in the characteristic of rating the values should be separated by using a space
- A slash (/) should be used for separation of values when referring to more than one voltage rating.
EXAMPLE: 240 and 480VAC = 240/480VAC

- A (TO) will be used to show all ranges.

EXAMPLE: 4 TO 12 AMPS = 4 TO 12A.

- When VAC and VDC are used together, always put the VAC first and use a comma to separate the two.

EXAMPLE: 120VAC and 20VDC = 120VAC, 20VDC 380VAC and 380VDC = 380VAC/DC

- AC should be added to the voltage when Hertz (HZ) or Phase (PH) is given.

EXAMPLE: 1,000V = 1KV 2,500V = 2.5KV 40,000A = 40KA

- When voltages of opposite polarities (positive/negative) are listed in the same characteristic value, use the following as an example.

EXAMPLE: CORRECT = +/-15VDC INCORRECT = +15VDC-15VDC

THREAD / CONNECTIONS

- GENERAL ORDER OF THREAD DESIGNATIONS ARE:

NUMBER OF THREADS PER INCH (PITCHES PER INCH), THREAD SERIES SYMBOL (e.g. UNC, UNRC, UNF, UNRF, UNEF, UNREF), THREAD CLASS SYMBOL (CLASS FIT), HAND (RIGHT-HAND or LEFT-HAND)

EXAMPLE: 20 UNC-2A LH

20 = NUMBER OF THREADS PER INCH (PITCHES PER INCH)

UNC = THREAD SERIES SYMBOL (UNIFIED COARSE THREAD)

2A = THREAD CLASS SYMBOL

LEFT HAND = LH

RIGHT HAND = RH

- "Compression, Flare, etc." are the connection value when describing a tube connection with a nut and/or ferrule.
- When referring to connection threads, they should be listed in the following sequence (when connection is not a separate characteristic): SIZE, MALE or FEMALE, INDUSTRY CONNECTION TYPE.

EXAMPLE: 1.5IN MNPT

- Make sure the sizes associate with the correct connection.

EXAMPLE: 0.5IN MNPT, 1IN FNPT

SIZE = 0.5IN, 1IN

CONNECTION = MNPT, FNPT

- If size and connection are separate characteristics, connection threads should be listed as MALE or FEMALE, INDUSTRY CONNECTION TYPE.

EXAMPLE: MNPT, FNPT

- NPTF will not be converted as FNPT. When connection indicates NPTF, the F does not mean "female" (NPTF is a Dry seal thread).

PRESSURE RATING

- Ratings are not converted and represented as-is found in the input description or from the public domains

EXAMPLE: CORRECT = 150LB / INCORRECT = Class 150 or 150PSI

- PSI is valid for gauges and set pressure ratings on relief valves.

- Will use the design pressure rating at a predetermined temperature (if available) in characteristic of Temperature Rating on valves.

EXAMPLE: 1600LB @ 600DEG F

Example of Standardization Rating:

<u>Source</u>	<u>Standardization</u>
0-30 #	0 TO 30PSI
0-30 PRSR	0 TO 30PSI

- Pipe Schedules may sometimes be used as a rating value (e.g. SCH 40, SCH 80, etc.)